

🔥 Error & Types

Def: Change of bits during transmission

- Single-bit → 1 bit change
- Burst → multiple bits

🔥 Circuit Switching

Def: Dedicated path before transmission

Phases:

- Setup
- Data Transfer
- Teardown

🔥 Packet Switching

Def: Data split into packets sent independently

🔥 Circuit vs Packet

Circuit	Packet
Dedicated path	No fixed path
Setup delay	Delay per packet

Circuit	Packet
Reliable	Flexible
Wastes bandwidth	Efficient

🔥 Datagram vs VC

Datagram	Virtual Circuit
No setup	Setup required
Diff path	Same path
Fast	Reliable

🔥 Static Routing

Def: Manual routing

- + Simple, Secure
- Not scalable, Manual

🔥 Distance Vector

Routers share tables with neighbors

🔥 Count to Infinity

Loop → distance increases infinitely

🔥 Link-State

Full network map + shortest path

🔥 DV vs Link-State

DV	Link-State
Slow	Fast
Loop issue	Less loops
Simple	Complex
Low resource	High resource

🔥 CSMA

Check channel before sending

🔥 CSMA/CD

Detect collision → resend

🔥 CSMA/CA

Avoid collision before sending

🔥 FDMA / TDMA

- FDMA → frequency division
- TDMA → time division